

AI-DRIVEN PERSONALIZED CANCER TREATMENT SUMMARY

PATIENT PROFILE

Name: Sarah Jenkins

Gender: female

Age: 34

MRN: MR-2026-5541-J

DOB: 22/11/1992

DOD: 15/1/2026

DISEASE CHARACTERIZATION & PATHOLOGY

Primary Diagnosis: Brain

Clinical Stage:

Histological Subtype:

WHO Grade: II

GENOMIC & VCF SUMMARY

Total Variants: 4

Actionable: 1

VUS:

Gene	Mutation Status	Clinical Impact	AI Weight
IDH1	Mutated	Favorable prognosis	High
MGMT	Methylated	Chemo-sensitive	High
IDH1	Mutated (48%)	Pathogenic (Clonal)	High

FUNCTIONAL & PERFORMANCE STATUS

KPS Score: 100

ECOG Score: 0

Functional Category: Independent

SYMPTOMS & COMORBIDITIES

Presenting Symptoms:

Comorbidities:

VOLUMETRIC ANALYSIS

Metric	Value	Metric	Value
Total Tumor Volume	33.22 cm ³	Active Core Volume	26.58 cm ³
Edema Volume	4.98 cm ³	Necrotic Fraction	1.66 %
Tumor Location	Left Frontal	Sphericity Index	

BIOMARKER ANALYSIS

Biomarker	Status	Clinical Meaning
IDH1	Mutated	Biologically less aggressive
MGMT	Methylated	Better response to Temozolomide
Resection Extent	Gross Total	Impacts residual disease risk
IDH1	Mutated	VCF detected significance: Pathogenic (Clonal)

AI-RECOMMENDED TREATMENT PLAN

{ "radiation": "Radiation therapy", "surgery": "Maximal safe resection", "systemic": "Chemotherapy (e.g., temozolomide)", "targeted": "Targeted therapy (e.g., IDH1 inhibitors) if applicable" }

Rationale:

- The patient has a localized brain tumor with an IDH1 mutation. The primary treatment goal is to maximize tumor resection while preserving neurological function, followed by radiation therapy and chemotherapy. Targeted therapy may be considered if the tumor is amenable to it.

OUTCOME PROJECTIONS

Overall Survival (Median): 15.2 Months
Quality of Life Score: 77.5/100

Progression-Free Survival: 9.1 Months
Primary Toxicity Risk: Monitor for fatigue and nausea